

$$V_U = N_{\text{ell}} V_8 \prod_{d=1}^8 (\Delta d_d \bar{v}_d) C_{\text{geom}} C, \quad V_8 = \pi^4/24, \quad C = C_{\text{bounds}} C_{\text{overlap}}, \quad \Delta d_d = 1 \text{ (normalized coordinates)}$$

